

Main Content

- **Iterative method**
- **Classical Time-Domain Method : Homogeneous solution + Particular solution (Similar to Continuous Systems)**
- **Zero-Input response + Zero-State response (Similar to Continuous Systems)**
Using convolutional sum to calculate the zero-state response of a system
- **Z-transform method $\rightarrow$ Inverse transform $\rightarrow y(n)$ (Chapter 6)**

1. Iterative method

- The difference equation itself is a recursive relationship
- But cannot obtain an analytical expression for the output sequence
- Suitable for using computers to solve problems

5. Difference Equations

Example 1

Given $y[n] = 0.9y[n-1] + u[n]$, $y[-1] = 0$, solve the equation

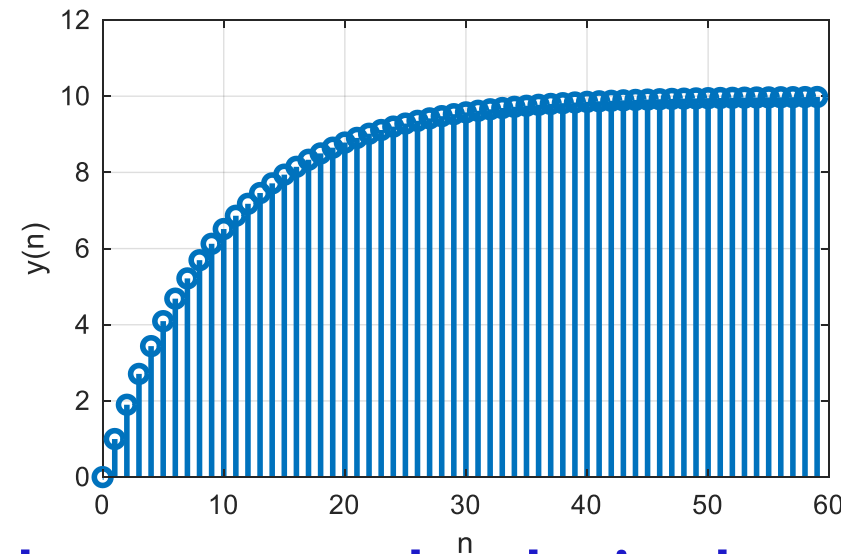
$$n=0 \quad y[0] = 0.9y[-1] + 1 = 1$$

$$n=1 \quad y[1] = 0.9y[0] + 1 = 1.9$$

$$n=2 \quad y[2] = 0.9y[1] + 1 = 2.71$$

$$n=3 \quad y[3] = 0.9y[2] + 1 = 3.439$$

...



From the recursive relationship, the output can be obtained

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1.9, 2.71, 3.439, \dots \right\}$$

Main Content

- Definition and Solution of the Unit Impulse Response for Discrete Time Systems
- Judging Causality and Stability from $h[n]$

1. Definition and Solution of the Unit Impulse Response

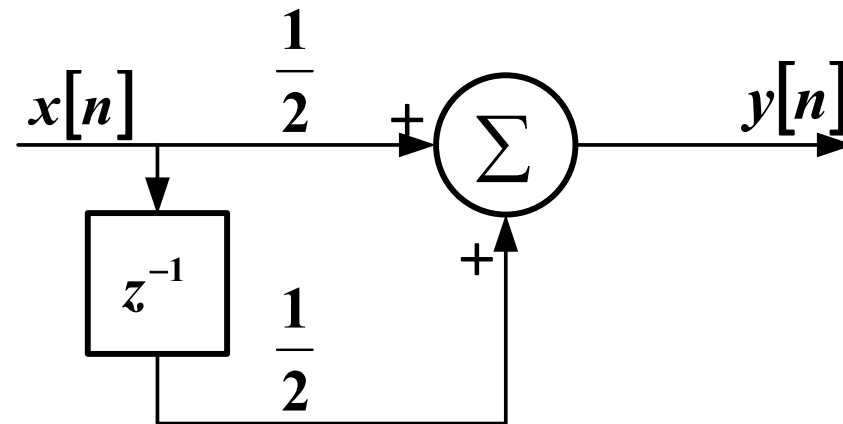


- Input $\delta[n]$
- **Zero-State:** $h[-n] = 0 \quad n = 1, 2, 3, \dots, N$

6. Unit Impulse Response

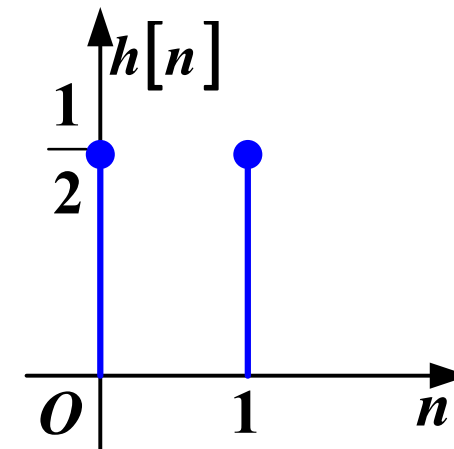
Example 1

Calculate the unit pulse response of the system



$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$$

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1]$$



6. Unit Impulse Response

Example 2

Find the unit pulse response of the **causal** system described by the following difference equation:

$$y[n] - 0.8y[n-1] = x[n]$$

$$h[n] - 0.8h[n-1] = \delta[n]$$

When $n > 0$

$$h[n] - 0.8h[n-1] = 0$$

Equation
becomes
homogeneous

Using the equivalent Initial condition method, the difference equation is regarded as a homogeneous equation $\delta(n)$ convert to the initial condition:

Zero-State condition

$$h[-1] = 0$$

Can iterate out $h[0]$

$$h[0] = 0.8h[-1] + \delta[0] = 1$$

6. Unit Impulse Response

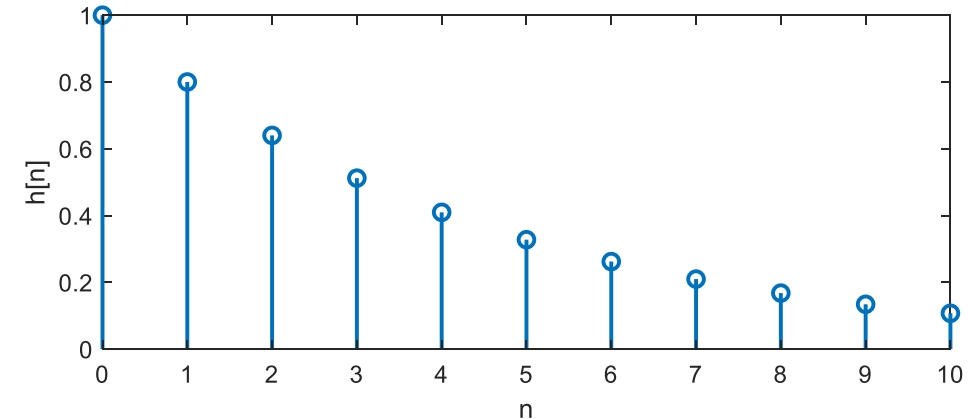
Example 2

$$h[n] - 0.8h[n-1] = 0$$

Characteristic equation $r - 0.8 = 0$

Characteristic root $r = 0.8$

Unit pulse response $h[n] = C0.8^n$



Substitute $h[0]=1$ into $h[n] = C0.8^n$, Get $C = 1$

Thus $h[n] = 0.8^n u[n]$

2. Judging Causality and Stability from $h[n]$

Causal system : output at any time n (or t) depends only on the current and past inputs, not future inputs

Necessary and sufficient conditions for a linear Time-invariant system to be a **causal system**:

$$n < 0 \quad h[n] = 0$$

The system meets the necessary and sufficient conditions for **stability**:

$$\sum_{n=-\infty}^{\infty} |h[n]| = P < \infty$$

The absolute sum of the unit impulse response is finite (**absolutely summable**)

Example 4

$$h[n] = a^n u[n]$$

(1) Causality:

One-sided, When $n < 0$, $h[n] = 0$

So, causal

(2) Stability:

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^{\infty} |a^n| = \begin{cases} \frac{1}{1-|a|} & |a| < 1 \\ \infty & |a| \geq 1 \end{cases}$$
$$|a| < 1, \quad h[n]$$

When $|a| < 1$, the system is stable

6. Unit Impulse Response

Summary



The **zero-state response** of the system under the excitation of $\delta[n]$ is expressed as $h[n]$

$$h[-n] = 0 \quad n = 1, 2, 3, \dots, N$$

Necessary and sufficient condition for a linear Time-invariant system to be **a causal system**:

$$n < 0 \quad h[n] = 0$$

Necessary and sufficient conditions for **stability**:

$$\sum_{n=-\infty}^{\infty} |h[n]| = P < \infty$$

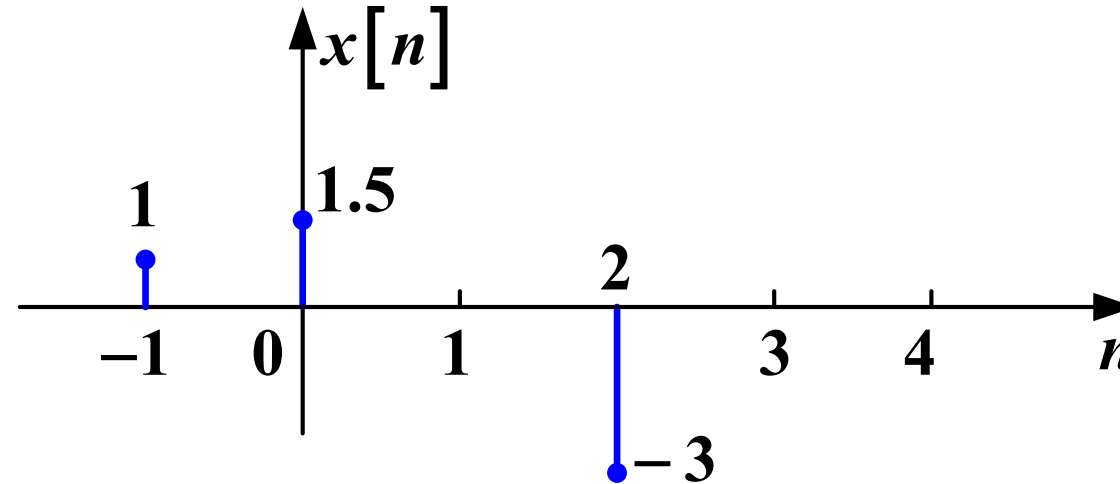
Main Content

- Definition of Convolution Sum
- Properties of Discrete Convolution
- Computation of Convolution Sum

1. Definition of Convolution Sum

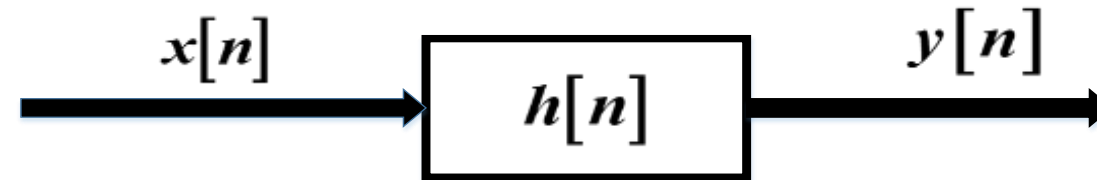
Any sequence $x[n]$ can be represented as a linear combination of weighted, shifted $\delta[n]$

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n-m]$$



$$x[n] = \delta[n+1] + 1.5\delta[n] - 3\delta[n-2]$$

1. Definition of Convolution Sum



Known $\delta[n] \rightarrow h[n]$

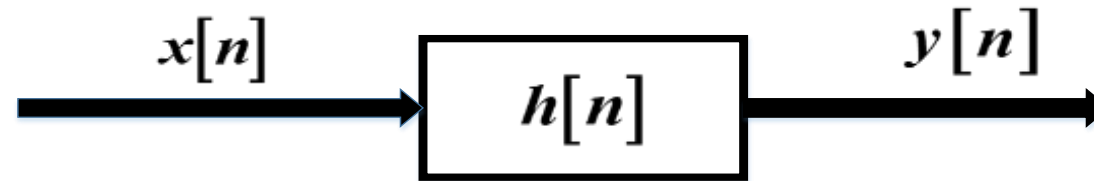
Time Invariant $\delta[n-m] \rightarrow h[n-m]$

Homogeneity $x[m]\delta[n-m] \rightarrow x[m]h[n-m]$

Additivity $x[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n-m]$

Output $y[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] = x[n] * h[n]$

1. Definition of Convolution Sum



$$y[n] = \sum_{m=-\infty}^{\infty} \textcolor{red}{x}[m] \textcolor{blue}{h}[n-m]$$
$$= \textcolor{red}{x}[n] * \textcolor{red}{h}[n]$$

$\textcolor{blue}{h}[n]$, linking input and output, i.e. **zero state response**

$$y[n] = x[n] * h[n]$$

2. Properties of Discrete Convolution

1. Commutative property

$$x[n] * h[n] = h[n] * x[n]$$

2. Associative property

$$x[n] * h_1[n] * h_2[n] = x[n] * [h_1[n] * h_2[n]]$$

3. Distributive property

$$x[n] * [h_1[n] + h_2[n]] = x[n] * h_1[n] + x[n] * h_2[n]$$

4. Convolution with impulse sequences

$$x[n] * \delta[n - m] = x[n - m]$$

3. Computation of Convolution Sum

$$x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]$$

The output range m is co-determined by $x[n]$ and $h[n]$

Discrete convolution process: sequence inversion → shift

→ multiplication → sum

(1) Analytical Method

(2) Graphical Method

(3) Use Properties

(4) Sum of Pairwise Multiplication

Example 1: Analytical Method

Given $h[n] = \alpha^n u[n]$ ($0 < \alpha < 1$), find the unit step response of the system

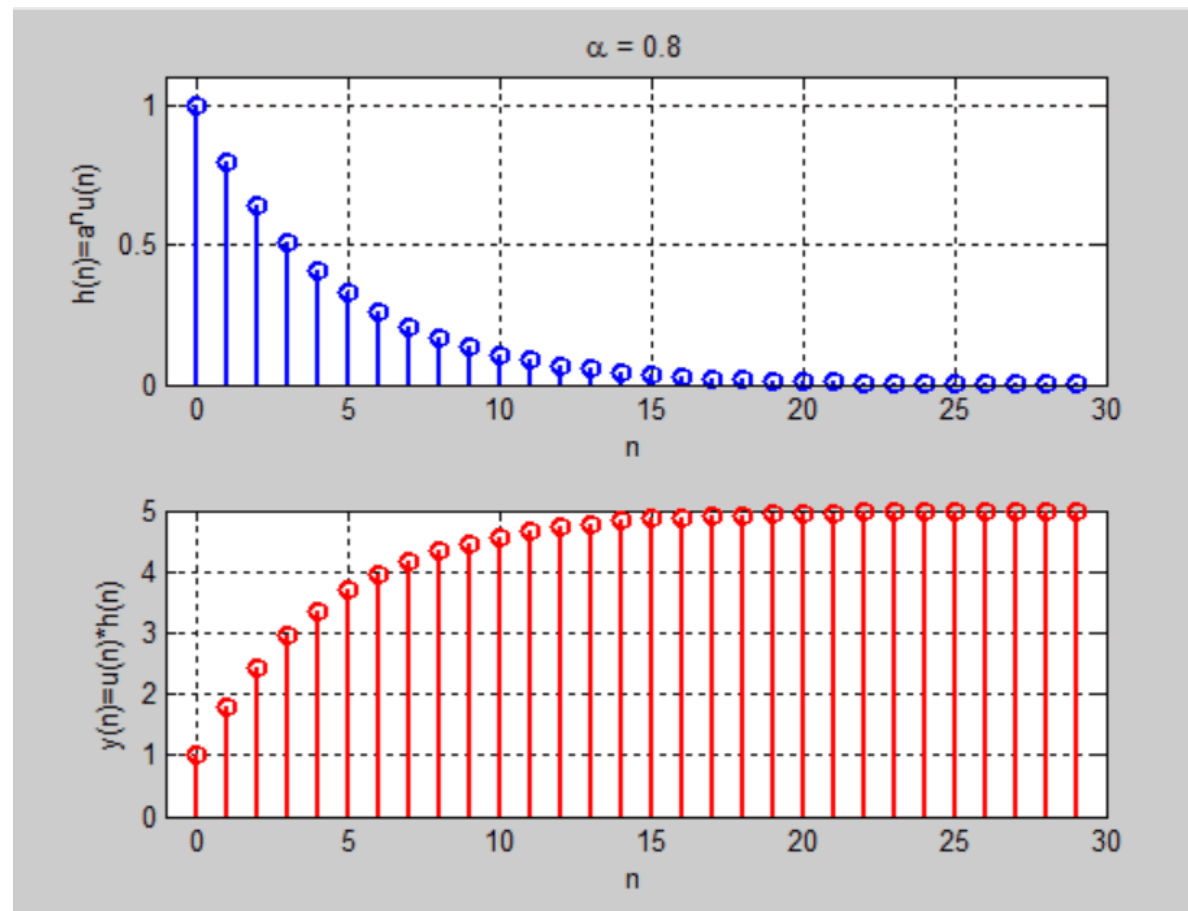
$$\begin{aligned} y[n] &= h[n] * u[n] \\ &= \sum_{m=-\infty}^{\infty} \alpha^m u[m] u[n-m] \quad \text{Argument : } \begin{cases} m \geq 0 \\ n-m \geq 0 \end{cases} \\ &= \left(\sum_{m=0}^n \alpha^m \right) \cdot u[n] \quad \Rightarrow \begin{cases} 0 \leq m \leq n \\ n \geq 0 \end{cases} \\ &= \frac{1 - \alpha^{n+1}}{1 - \alpha} u[n] \end{aligned}$$

7. Discrete Convolution

Example 1: Analytical Method

$$h[n] = \alpha^n u[n] \quad (0 < \alpha < 1) \text{ Input } u[n]$$

$$y[n] = \frac{1 - \alpha^{n+1}}{1 - \alpha} u[n]$$



Example 2: Graphical Method, Multiply and Sum

Consider the following finite length signal

$$x[n] = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{other } n \end{cases}$$

Calculate $y[n] = x[n] * x[n]$ using analytical methods

Analysis:

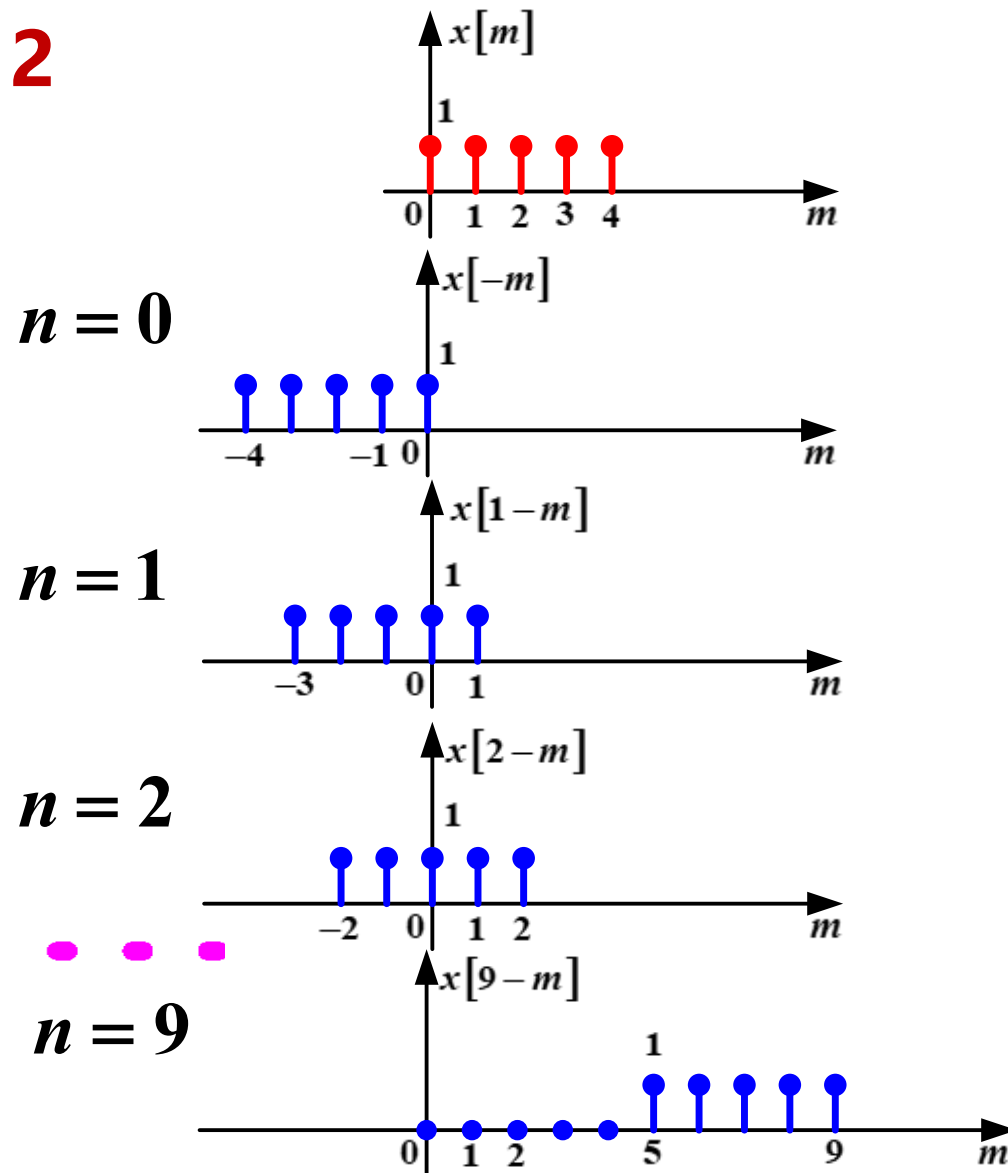
$$x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]$$

Discrete convolution process: sequence inversion → shift → multiplication → sum

The m range is co-determined by $x[n]$ and $h[n]$

7. Discrete Convolution

Example 2



$$y[0] = 1 \times 1 = 1$$

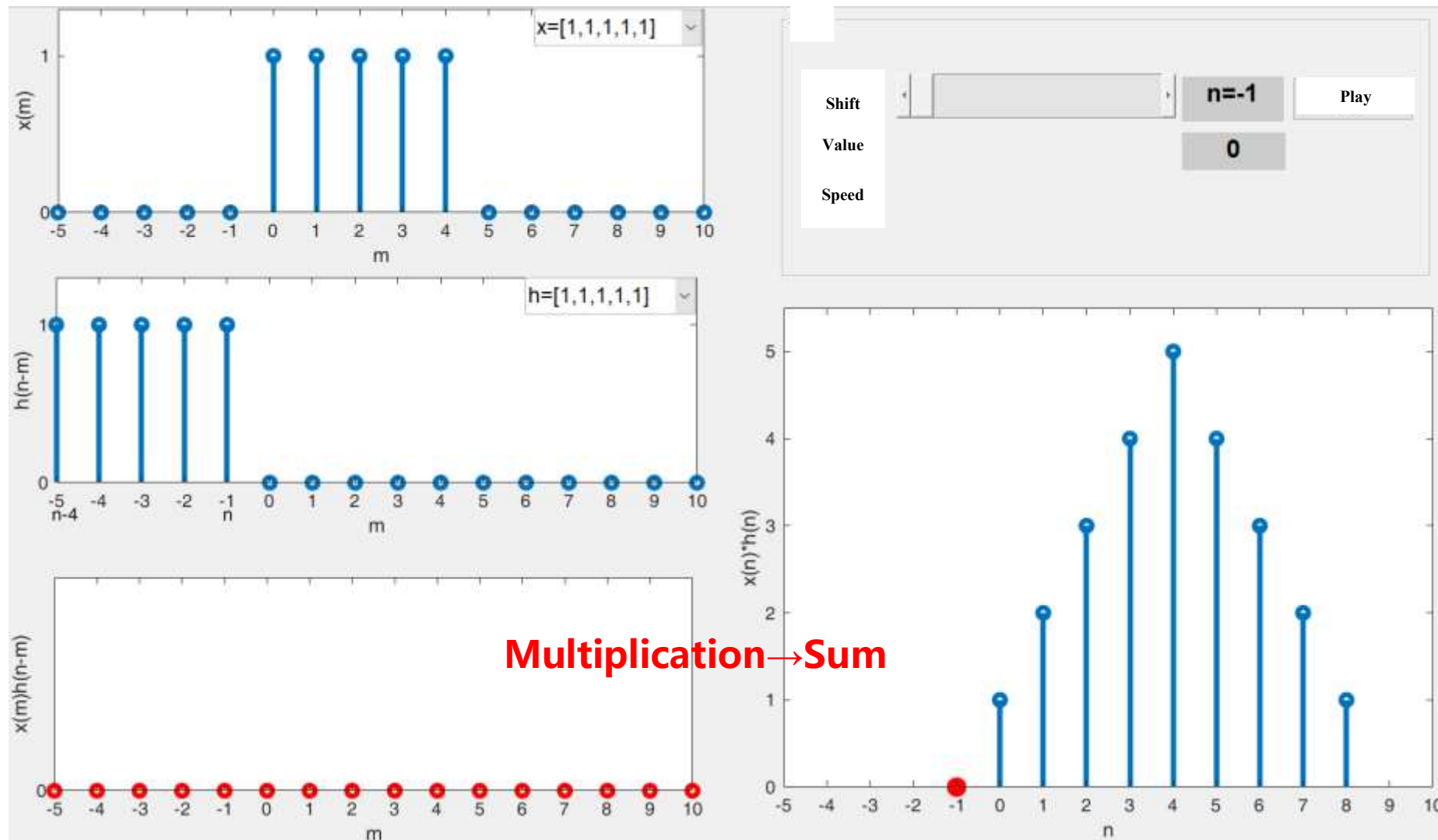
$$y[1] = 1 \times 1 + 1 \times 1 = 2$$

$$y[2] = 3$$

$$y[9] = 0$$

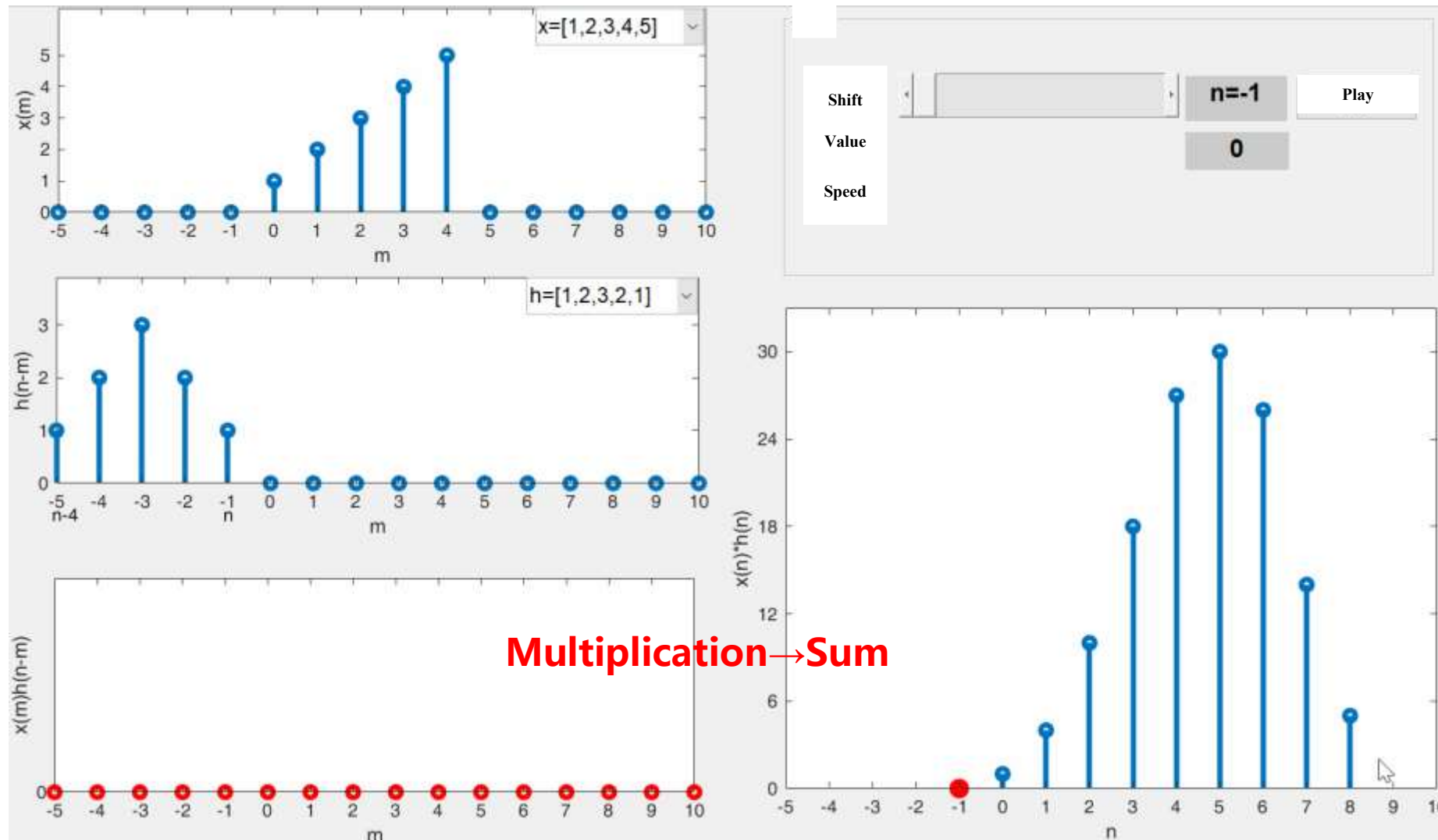
7. Discrete Convolution

Example 2: Graphical Method, Multiply and Sum



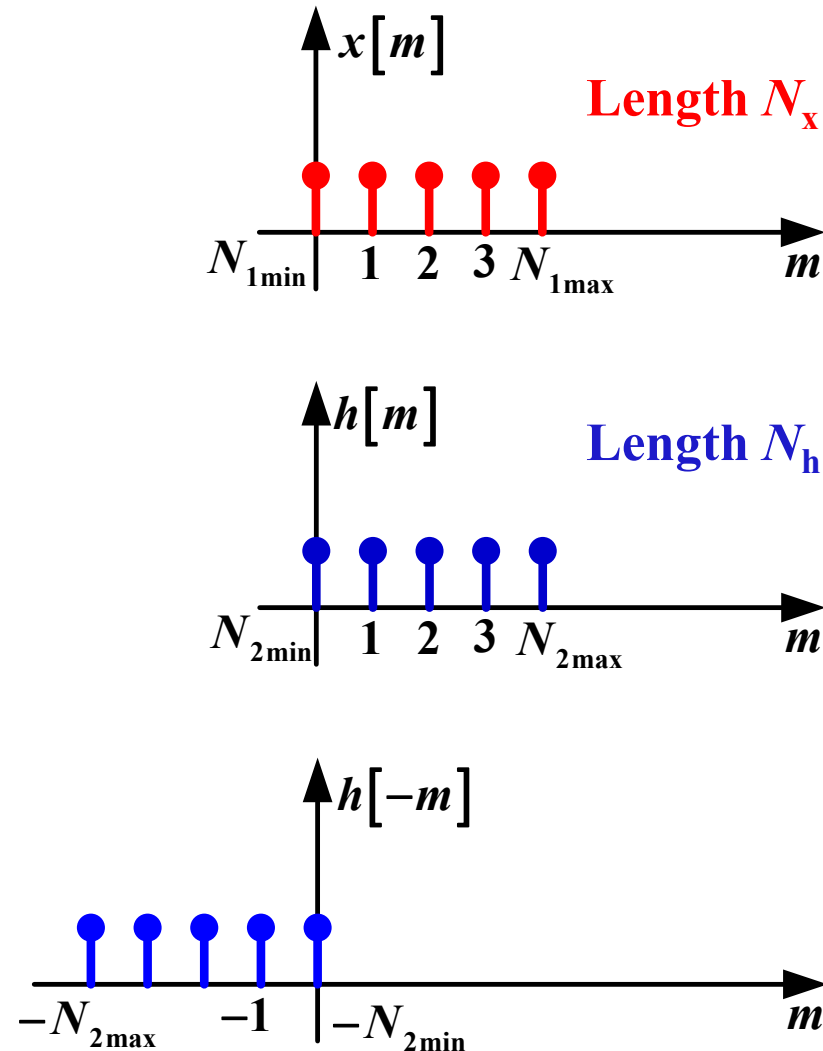
7. Discrete Convolution

Example 2: Graphical Method, Multiply and Sum



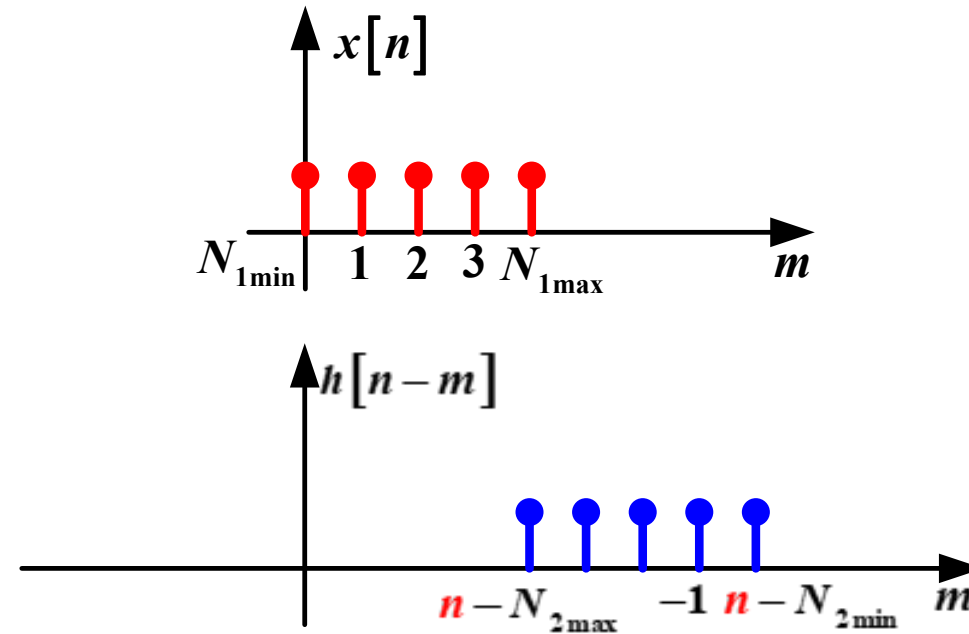
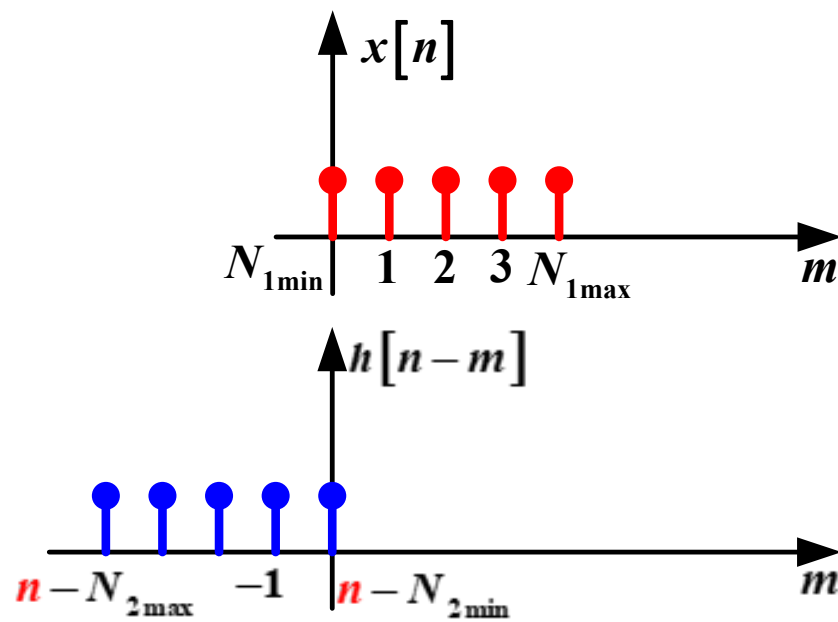
7. Discrete Convolution

Determining the Range of Convolution Results



7. Discrete Convolution

Determining the Range of Convolution Results



Lower limit 1 + Lower limit 2 $\leq n \leq$ Upper limit 1 + Upper limit 2

The length of convolutional signals $N_x + N_h - 1$

Example 3: Use Properties

$$\text{Given } x[n] = R_3[n], h[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3 \right\}, \quad \text{Get } x[n] * h[n].$$

Analysis:

$$x[n] * \delta[n-m] = x[n-m]$$

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

$$h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

Example 3: Use Properties

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

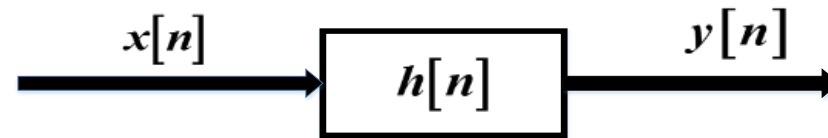
$$h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

Distributive property

$$\begin{aligned} x[n] * h[n] &= \delta[n] + 2\delta[n-1] + 3\delta[n-2] \\ &\quad + \delta[n-1] + 2\delta[n-2] + 3\delta[n-3] \\ &\quad + \delta[n-2] + 2\delta[n-3] + 3\delta[n-4] \\ &= \delta[n] + 3\delta[n-1] + 6\delta[n-2] + 5\delta[n-3] + 3\delta[n-4] \end{aligned}$$

Summary

- Definition of Convolution Sum
- Properties of Discrete Convolution
- Computation of Convolution Sum



$$y[n] = \sum_{m=-\infty}^{\infty} \textcolor{red}{x}[m] \textcolor{blue}{h}[n-m] = x[n] * h[n]$$

- (1) Analytical Method
- (2) Graphical Method
- (3) Use Properties
- (4) Sum of Pairwise Multiplication



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

